ALLOCATION OF LAND WITH LINEAR PROGRAMMING

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ABSTRACT
We formulate a linear program for allocating land among crops and study random yields with a simple stochastic model. The optimal plan is found with the simplex method.
KEYWORDS: linear programming; stochastic yield

1 INTRODUCTION

Agriculture is important for food security (World Bank, 2014). The planning problem was described by Food and Agriculture Organization of the United Nations (2023) and solved by Smith and Doe (2020).

2 METHODS

Let the decision variables be defined as follows.

$$x_1 = \text{allocated land for Coffee (hectares)} \tag{1}$$

The objective is

$$Z = p_1x_1 + p_2x_2 + p_3x_3 + p_4x_4 + p_5x_5 + p_6x_6 \tag{2}$$

subject to the land constraint

$$x_1 + x_2 + x_3 \leq \lambda \tag{3}$$

and the yield follows $Y_i = \mu_i + \epsilon_i$ with

$$\epsilon_i \sim N(0, \sigma_i^2), \quad \text{with } \sigma_i \text{ as its standard deviation} \tag{4}$$

The implied area is

$$a_i = \frac{t_i}{\mu_i} \tag{5}$$

Table 1: Yield per hectare by crop

Crops	Yield (tonnes/ha)	Cost (PGK/ha)
Coffee	0.75	498
Sweet Potato	14.00	2,500
Banana	6.50	1,776.5

3 CONCLUSION

The model is useful for planning.

REFERENCES

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